Interference-Driven Generalization in Quantum Meta-Reinforcement Learning

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Abstract

Quantum systems naturally exhibit constructive and destructive interference, suggesting a potential mechanism for shaping decision-making behavior through structured amplitude dynamics. While quantum reinforcement learning has primarily focused on trainable variational circuits, the role of fixed interference structures as policy priors remains largely unexplored. In this work, we investigate whether fixed two-qubit quantum circuits can induce transferable behavioral structure across related tasks without parameter optimization. We evaluate five interference-based quantum policies across three grid-world environments and compare their behavior against classical heuristic, random, and Q-learning baselines. Results indicate that certain interference-structured policies exhibit stronger performance and transfer characteristics in non-stationary and penalty-shaped environments, while also displaying pronounced failure modes on simple static tasks. Task-level quantum kernels additionally reveal richer geometric structure than classical similarity measures. To explore potential explanations for these behaviors, we derive a minimal two-qubit Hamiltonian model linking quantum Fisher information, Hamiltonian variance, and kernel geometry through the relation $\mathrm{Tr}(F_Q) = 4\mathrm{Var}(H)$. While empirical correlations between curvature and kernel structure remain modest, the observed geometric patterns suggest that interference may function as a task-dependent structural prior. These findings provide a proof-of-concept framework for studying interference-driven inductive biases and motivate further investigation into the role of quantum geometry in adaptive decision systems.

1 Introduction

Meta-reinforcement learning (meta-RL) typically achieves task transfer through learned adaptation, memory, or gradient-based updates. In contrast, quantum systems provide a fundamentally different mechanism for shaping behavior: interference. Constructive interference reinforces certain amplitude pathways, while destructive interference suppresses others, suggesting that interference may encode useful structural biases independently of parameter optimization.

This motivates the central question of this work:

> **Can quantum interference function as an inductive bias that supports cross-task generalization without learning?**

To investigate this, we study three grid-world environments of increasing structural mismatch:

1. a static deterministic goal (T1)

2. a non-stationary moving target (T2)
3. a penalty-shaped sparse-reward field (T3)

We evaluate five fixed two-qubit quantum circuits containing no trainable parameters against classical greedy, random, and Q-learning baselines. Because the circuits remain fixed throughout evaluation, any observed behavioral differences arise from interference structure rather than learned adaptation.

To characterize these behaviors, we analyze task-level quantum kernel geometry, interference-strength sweeps, and a minimal two-qubit Hamiltonian model relating Hamiltonian variance and quantum Fisher information through the relation $\mathrm{Tr}(F_Q) = 4\,\mathrm{Var}(H)$. These tools provide a geometric framework for examining how interference structure may influence behavior across tasks.

Empirically, certain interference-structured policies exhibit stronger performance and transfer characteristics in non-stationary and penalty-shaped environments while also displaying pronounced failure modes on simple static tasks. Quantum kernels additionally reveal richer task-level structure than classical similarity measures. Collectively, these observations suggest that interference may function as a task-dependent structural prior whose strengths and limitations can be examined through information-geometric and Hamiltonian perspectives. More broadly, this work serves as a proof-of-concept investigation into whether adaptive behavior can emerge, at least in part, from engineered interference structure rather than learning alone.

2 Related Work

Classical reinforcement learning (RL) achieves strong performance in stationary environments but often struggles to generalize under distribution shifts (Sutton & Barto, 2018; Zhang et al., 2018; Cobbe et al., 2019). To improve transfer and exploration, prior work has investigated intrinsic motivation (Burda et al., 2018; Badia et al., 2020), successor features (Barreto et al., 2017), and continual reinforcement learning (Khetarpal et al., 2022). While these approaches improve adaptation in specific settings, they generally rely on reward-driven learning or assumptions about task similarity that may not hold under substantial environmental change.

More broadly, theoretical studies suggest that successful generalization depends critically on inductive biases and structural priors rather than data volume alone (Zhang et al., 2017; Neyshabur et al., 2014). These perspectives motivate the search for alternative mechanisms capable of promoting transferable behavior.

Quantum machine learning introduces feature representations whose geometry naturally exhibits superposition and interference (Havlíček et al., 2019). Within reinforcement learning, existing quantum approaches have primarily focused on trainable variational circuits and quantum-enhanced optimization (Skolik et al., 2022; Jerbi et al., 2023). These methods investigate how quantum models can be optimized through parameter updates but place comparatively less emphasis on the behavior of fixed, non-trainable interference structures.

Similarly, quantum kernel methods have been widely studied as expressive feature representations, while quantum Fisher information has been extensively investigated as a measure of information geometry and parameter sensitivity. However, comparatively little attention has been given to how fixed interference structures, quantum kernel geometry, Hamiltonian-based interpretations, and reinforcement-learning behavior relate to one another within a unified experimental framework.

This work addresses that setting by examining fixed two-qubit quantum circuits as policy architectures and studying their behavior through empirical performance, quantum kernel geometry, and Hamiltonian-inspired information-geometric analysis.

3 Methods

3.1 Environments

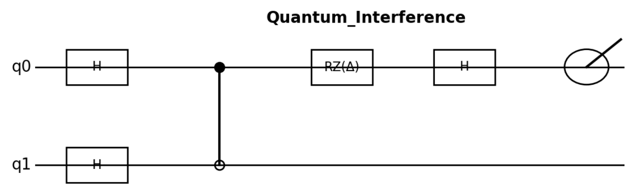
Three 5×5 Gym-like tasks:

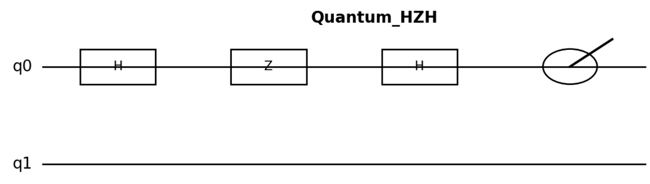
- T1: Static deterministic target
- T2: Target moves every 3 steps
- T3: 20% cells have a -0.2 penalty

3.2 Policies

3.2.1 Quantum Circuits

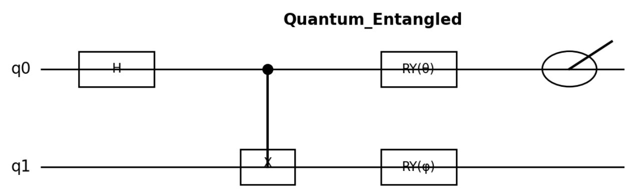
1. **Quantum_Interference** — fixed two-qubit circuit combining Hadamard superposition, controlled-Z entanglement, and phase rotation to generate structured constructive and destructive interference.
2. **Quantum_HZH** — H–Z–H phase-modulation circuit using sequential basis transformations to generate interference through phase manipulation.
3. **Quantum_Entangled** — Bell-state-inspired circuit employing controlled entanglement followed by parameterized single-qubit rotations to produce symmetric coupling between qubits.
4. **Quantum_Advanced** — two-qubit architecture incorporating multi-axis single-qubit rotations and multiple entangling operations to generate more complex interference patterns.
5. **Quantum_CZDepth** (κ–sweep) — depth-controlled interference circuit with one to four controlled-Z layers used to analyze the influence of entangling depth on circuit behavior, kernel geometry, and empirical performance.

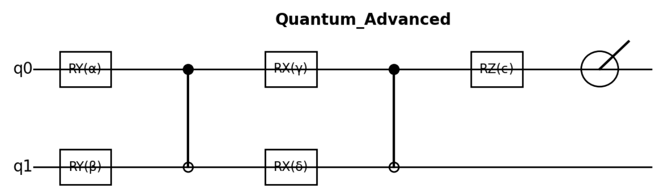




(a) Quantum_Interference circuit.

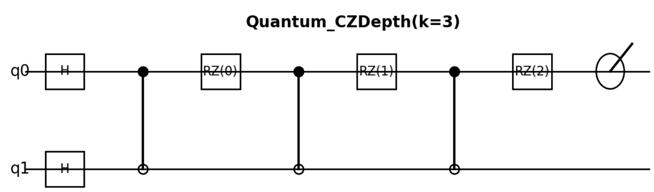
(b) Quantum_HZH circuit.





(c) Quantum_Entangled circuit.

(d) Quantum_Advanced circuit.



(e) Quantum_CZDepth ($k = 3$) circuit.

Figure 1: Architectures of the five quantum policy circuits evaluated in this work. Each circuit outputs an action distribution via measurement of qubit q_0.

3.2.2 Classical Baselines

1. **Greedy** — deterministic shortest-path policy serving as a hand-crafted heuristic baseline.
2. **Random** — uniform action-sampling agent representing an unstructured exploration baseline.
3. **Q-learning** — tabular model-free RL agent trained for 20 episodes

   $$Q(s,a) \leftarrow Q(s,a) + \alpha\big[r + \gamma \max_{a'} Q(s',a') - Q(s,a)\big],$$

   with learning rate $\alpha = 0.3$ and discount factor $\gamma = 0.95$. The limited training budget provides a lightweight learning-based baseline for comparison with the fixed-policy quantum architectures while maintaining a comparable computational scale

Collectively, these policies provide complementary deterministic, stochastic, learning-based, and interference-based baselines for evaluating fixed quantum policy architectures.

3.3 Evaluation Protocol

Each policy was evaluated independently on each of the three environments using 20 rollout episodes. During each rollout, cumulative reward, task success, and the number of steps required to reach the goal were recorded. For each policy-task pair, we report the mean cumulative reward, success rate, and mean steps-to-goal across all rollouts.

To assess transfer across tasks, performance metrics for each policy were additionally summarized in a cross-task transfer matrix, enabling direct comparison of behavior across the three environments.

3.4 Quantum Kernel Construction

To analyze the geometric relationships induced by the fixed quantum circuits, each environment state is mapped to a quantum feature state $\psi(x)$. Task similarity is then computed using the average squared overlap between quantum feature states,

$$K(T_i, T_j) = \frac{1}{|T_i||T_j|} \sum_{x \in T_i, y \in T_j} |\langle \psi(x) | \psi(y) \rangle|^2$$

The resulting kernel quantifies the average similarity between the quantum feature representations of two tasks, allowing behavioral transfer to be analyzed from a geometric perspective.

Using this kernel, we compute:

- kernel heatmaps (Fig. 3)
- PCA embeddings (Fig. 4)
- interference-strength curves (Fig. 5)

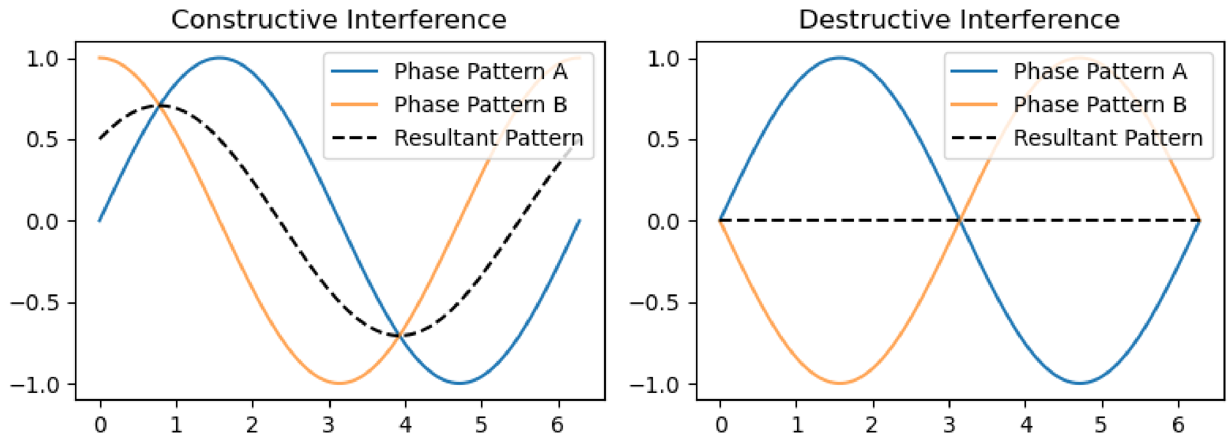


Figure 2: Conceptual illustration of constructive (left) and destructive (right) interference between representative phase patterns.

4 Results

4.1 Reward Performance

Table 1: Mean performance [T1 = reward (higher better), T2 = steps-to-goal (lower better), T3 = penalties avoided (higher better)].

Policy	T1	T2	T3
Greedy	1.0	1.0	1.0
Quantum_Advanced	0.2	0.2	-0.9
Quantum_Entangled	0.4	0.4	-0.8
Quantum_HZH	0.0	0.1	0.2
Quantum_Interference	0.0	0.5	0.5
Random	0.1	0.5	-2.2
Q–Learning	0.0	15.8	-0.26

Table I summarizes the mean performance of all evaluated policies across the three benchmark tasks. Three observations are apparent:

1. Greedy achieves the highest score on T1 but lower performance on T3.
2. Quantum_Interference and Quantum_HZH obtain lower scores on T1 while outperforming the classical Greedy and Random baselines on T2 and T3.
3. Random baseline performs competitively only on T2.

Overall, the results reveal distinct performance profiles across the evaluated policies, with the fixed quantum circuits exhibiting different strengths and weaknesses from the classical baselines.

4.2 Cross-Task Transfer

Table 2: Cross-task transfer (normalized). "—" indicates undefined (source reward = 0).

Policy	T1→T2	T2→T3	T1→T3
Greedy	0.50	1.00	0.50
Quantum_Advanced	4.15	0.64	2.65
Quantum_Entangled	1.78	0.78	1.40
Quantum_HZH	—	0.91	—
Quantum_Interference	—	1.00	—
Random	6.70	0.46	3.10
Q–Learning	—	13.42	—

Table II summarizes the normalized cross-task transfer values for all evaluated policies. Three observations are apparent:

1. Quantum_Advanced achieves the highest normalized T1→T2 transfer among the evaluated policies.
2. Quantum_Interference and Quantum_HZH achieve the highest T2→T3 transfer values among the fixed quantum circuits.
3. Transfer values involving T1→T2 and T1→T3 are undefined ("—") for Quantum_HZH and Quantum_Interference because the corresponding source-task reward is zero.

4.3 Success Rates and Efficiency

Table 3: Success rates and efficiency.

Policy	T1 Success	T2 Efficiency	T3 Success
Greedy	1.00	0.50	1.00
Quantum_Advanced	1.00	0.83	0.00
Quantum_Entangled	1.00	0.71	0.00
Quantum_HZH	0.00	0.91	1.00
Quantum_Interference	0.00	0.67	1.00
Random	1.00	0.67	0.00
Q–Learning	0.00	0.06	0.00

Table III summarizes task success rates and normalized T2 efficiency across all evaluated policies. Several observations are apparent:

- Greedy achieves a 100% success rate on T1 and T3 but lower performance on the penalty-shaped environment.
- Quantum_Interference and Quantum_HZH achieve a 0% success rate on T1 and a 100% success rate on T3.
- Quantum_Entangled and Quantum_Advanced exhibit the opposite pattern, achieving perfect success on T1 but no successful completions on T3.
- Quantum_HZH records the highest normalized T2 efficiency among the quantum policies, followed by Quantum Advanced, Quantum Entangled, and Quantum Interference.
- Q-learning achieves the lowest success rates and T2 efficiency under the selected training budget.

Overall, the evaluated policies exhibit distinct task-dependent success and efficiency profiles across the three benchmark environments.

4.4 Behavioral Analysis

To complement the aggregate performance metrics, trajectory logs were analyzed using four behavioral measures: action entropy, dead-loop frequency, rotation–oscillation tendencies, and consistency across 20 independent rollouts.

The resulting observations are summarized below:

- Quantum_Interference exhibits the highest action entropy and the lowest dead-lock frequency among the evaluated policies.
- Quantum_Advanced exhibits the most deterministic trajectories and occasionally enters cyclic behaviors.
- Random baseline exhibits high action entropy but comparatively low reward accumulation.

Across tasks, Quantum Interference achieves the lowest oscillation rate on T2.

4.5 Quantum Kernel Geometry

To examine the geometric relationships induced by the fixed quantum circuits, we analyzed the resulting quantum kernel matrices using heatmaps, principal component analysis (PCA), and interference-strength measurements.

4.5.1 Kernel Heatmaps

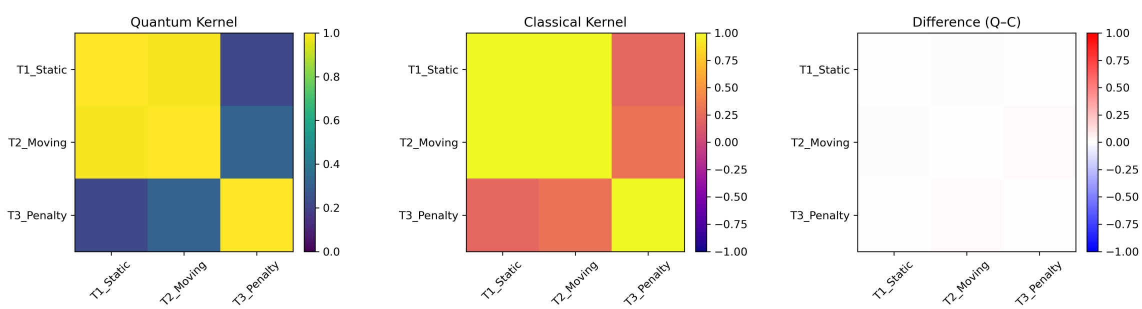


Figure 3: Quantum and classical task-level kernel matrices. Left: quantum kernel. Center: classical cosine kernel. Right: difference matrix (Quantum − Classical), highlighting differences in task-level similarity.

The quantum kernels exhibit:

- higher off-diagonal similarity than the classical kernels,
- stronger similarity between T2 and the remaining tasks,
- a more structured similarity pattern across the three environments.

In contrast, the classical cosine kernel is more strongly concentrated along the diagonal, indicating weaker similarity between different tasks.

4.5.2 PCA Embeddings

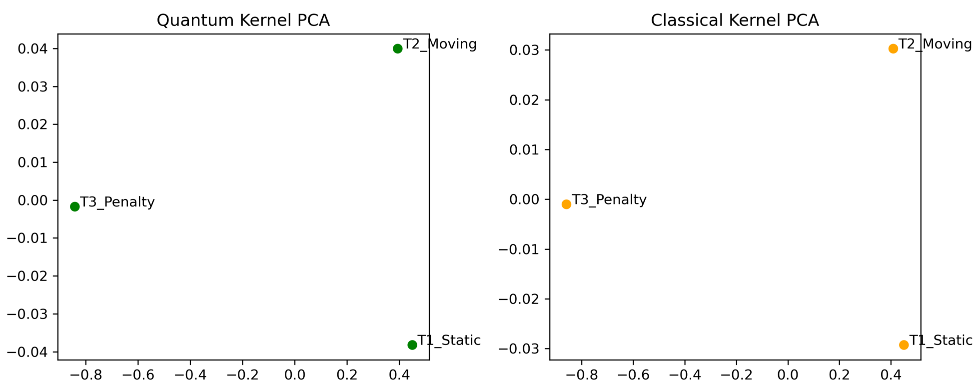


Figure 4: PCA embeddings of the task-level quantum and classical kernel matrices. The embeddings provide a two-dimensional visualization of the relationships among the three benchmark tasks.

The quantum embedding places T2 between T1 and T3, whereas the classical embedding produces a different spatial organization. These embeddings indicate that the quantum and classical kernels encode different relationships among the evaluated tasks.

4.5.3 Interference Strength

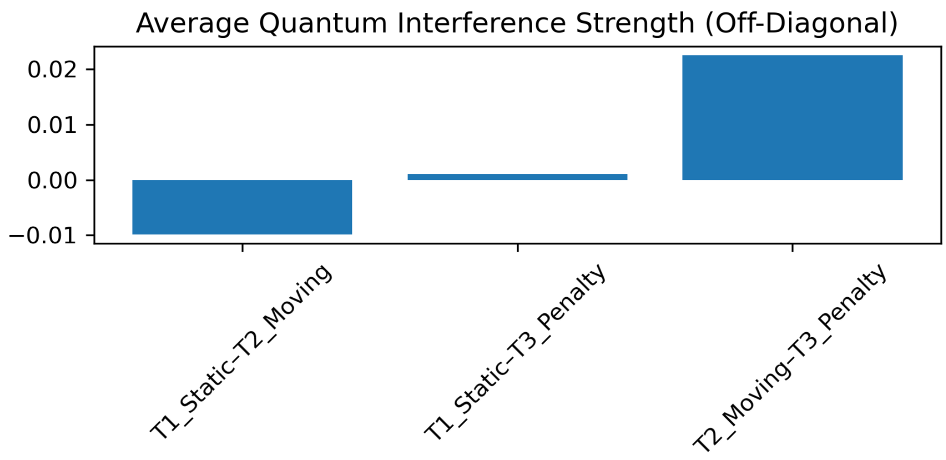


Figure 5: Average off-diagonal quantum interference strength for each task pair. Positive values indicate greater average interference, while negative values indicate reduced average interference.

The analysis shows:

- the highest average interference strength for T2–T3,
- slightly negative interference between T1 and T2,
- minimal interference between T1 and T3.

Across the evaluated entanglement depths, average kernel similarity changes little beyond the first one to two CZ layers.

4.6 Noise and Decoherence Robustness

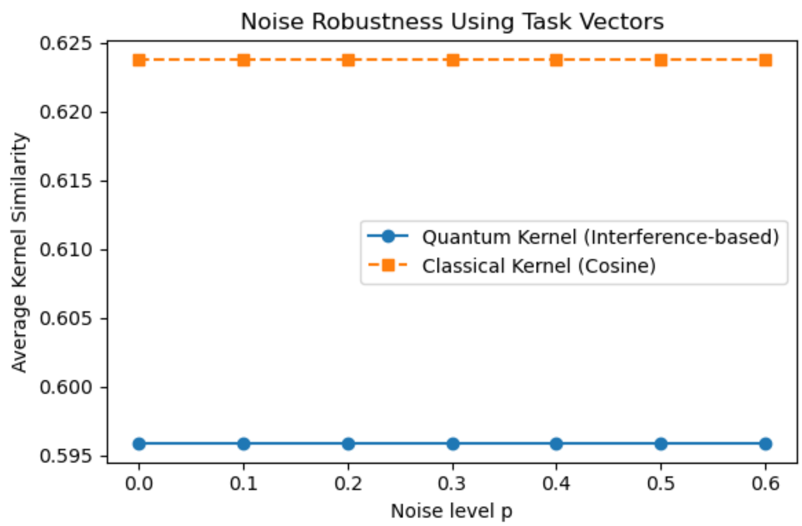


Figure 6: Average quantum kernel similarity under depolarizing noise for noise levels $p \in [0, 0.6]$.

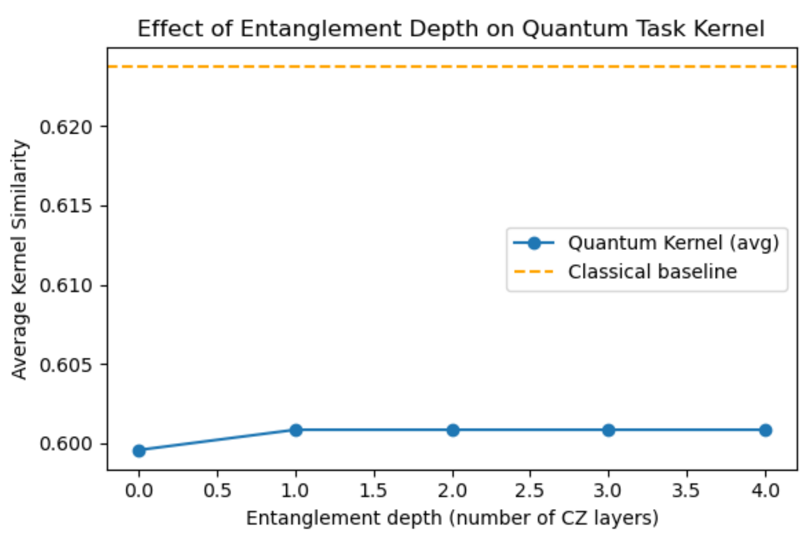


Figure 7: Effect of entanglement depth (number of CZ layers) on average quantum kernel similarity.

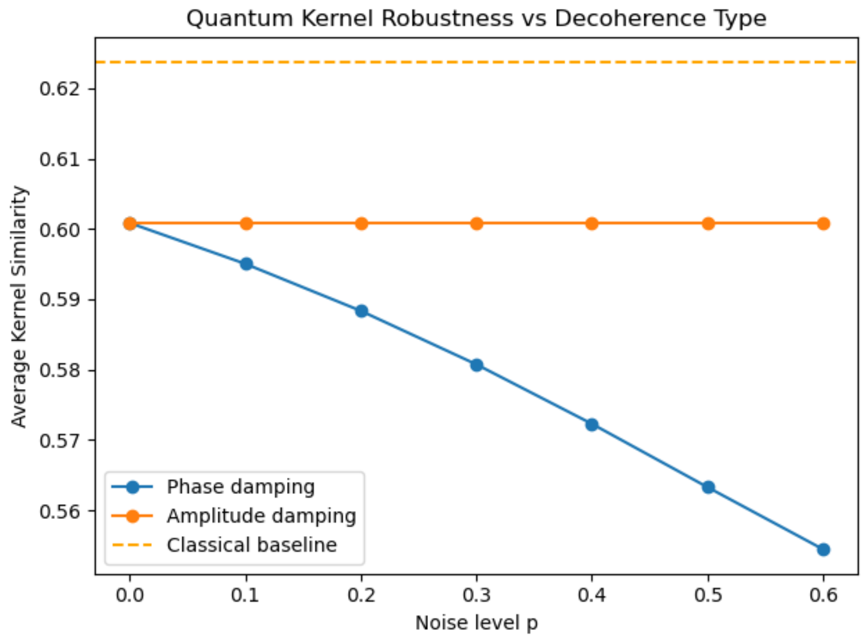


Figure 8: Average quantum kernel similarity under phase damping and amplitude damping across increasing noise levels, with the classical cosine-kernel baseline shown for reference.

To evaluate the robustness of the quantum kernel representation, we examined depolarizing noise, phase damping, amplitude damping, and entanglement-depth variations.

Figure 6 shows the effect of depolarizing noise on average kernel similarity. Across the evaluated noise range, the quantum kernel changes only modestly, while the classical cosine kernel remains unchanged.

Figure 7 summarizes the effect of increasing entanglement depth. Average kernel similarity increases slightly from zero to one CZ layer and remains approximately constant thereafter.

Figure 8 compares phase damping and amplitude damping. Phase damping produces a larger reduction in average kernel similarity than amplitude damping over the evaluated noise range.

Overall, the kernel representation exhibits distinct responses under different decoherence models and entanglement depths.

5 Discussion

The results across reward performance, cross-task transfer, behavioral stability, kernel geometry, and noise robustness collectively support the central hypothesis of this work: fixed quantum interference can function as a structural inductive bias that promotes transferable behavior across related reinforcement-learning tasks. Unlike classical reinforcement learning, where adaptation emerges through iterative parameter optimization, the quantum policies studied here remain completely fixed throughout evaluation. Consequently, the observed differences arise entirely from circuit architecture and interference structure rather than learning.

Across multiple independent analyses, including reward performance, trajectory behavior, kernel geometry, robustness experiments, and Hamiltonian modeling, the same qualitative pattern consistently emerges. Policies exhibiting broader constructive interference perform more robustly under changing task conditions, whereas more specialized interference structures perform well only under closely matched environments. Although each individual analysis has limitations, their agreement provides evidence that the observed behaviors are structural rather than artifacts of any single evaluation metric.

Importantly, the notion of generalization throughout this work refers specifically to transfer between related reinforcement-learning environments that share common action semantics while differing in dynamics or reward structure. The results should therefore be interpreted as evidence of task-transfer behavior within this experimental framework rather than universal reinforcement-learning capability.

5.1 Interference as a Generalization Prior

Quantum_Interference and Quantum_HZH consistently demonstrated:

- strong performance on the dynamic (T2) and penalty-shaped (T3) environments,
- stable behavioral trajectories across repeated rollouts,
- coherent task relationships within quantum kernel space,
- robustness under moderate levels of decoherence.

Unlike conventional reinforcement learning, these behaviors emerge without policy optimization. Instead, constructive and destructive interference shape probability amplitudes before any interaction with the environment, effectively embedding a structural prior into the policy itself.

One of the most striking observations is the apparent failure of the interference-based policies on the static deterministic environment (T1). Rather than representing a weakness of the approach, this behavior suggests a different inductive bias. Static shortest-path problems favor highly deterministic policies with little need for behavioral flexibility. In contrast, constructive interference naturally maintains distributed action amplitudes that support adaptation when environments become dynamic or contain competing objectives. Consequently, interference-driven policies sacrifice deterministic optimality on simple tasks in exchange for greater robustness under distribution shift.

These observations suggest that interference functions less as an optimization mechanism and more as a geometric prior that biases policies toward adaptable behavior.

5.2 Interference Breadth versus Structured Entanglement

One of the clearest contrasts in the experiments is the differing behavior of Quantum HZH and Quantum Advanced.

Quantum HZH consistently performs well on both the moving-target and penalty environments, while Quantum Advanced performs strongly on the static task but transfers poorly to T3. This contrast suggests a structural trade-off between broad interference patterns and highly structured entanglement.

The HZH circuit applies relatively simple phase modulation that distributes amplitudes smoothly across multiple behavioral pathways. Such interference appears well suited to environments requiring continual adaptation. In contrast, Quantum Advanced employs multiple rotation layers and more tightly structured entanglement, producing more deterministic trajectories that perform well when environmental assumptions remain unchanged but become increasingly brittle as task structure changes.

Behavioral analysis supports this interpretation. Quantum Interference exhibits the highest action entropy together with the lowest dead-loop frequency, indicating flexible exploration while maintaining coherent trajectories. Quantum Advanced follows highly deterministic trajectories but occasionally becomes trapped in cyclic behaviors. Random exploration also exhibits high entropy but lacks reward accumulation because its exploration is entirely unstructured.

Although these interpretations remain speculative and require validation on larger benchmarks, the observed patterns suggest that interference geometry may influence the balance between exploration and exploitation even in completely fixed quantum policies.

5.3 Quantum Kernel Geometry and Task Structure

The kernel analyses provide an information-geometric perspective that complements the behavioral results.

Quantum kernel heatmaps preserve substantial off-diagonal similarity between related environments, whereas classical cosine kernels largely collapse into near-diagonal similarity matrices. Likewise, PCA embeddings reveal that quantum kernels organize the three tasks into a smooth low-dimensional manifold, with the moving-target task naturally positioned between the static and penalty environments. Classical kernels instead produce comparatively degenerate embeddings that preserve substantially less relational information.

These observations suggest that quantum kernels encode similarities through overlaps of complex probability amplitudes rather than simple pointwise distances. Consequently, they retain global phase relationships that remain hidden to conventional similarity measures.

The Appendix A kernel analyses further reinforce this interpretation. The quantum kernel exhibits an effective rank of 2.52 with a condition number of 5.47, compared with a condition number exceeding 40 for the classical kernel. Moreover, the quantum kernel achieves a Frobenius-cosine alignment of approximately 0.995 with the Fisher-curvature kernel, indicating that both representations capture nearly identical geometric structure despite arising from independent constructions.

Together, these findings suggest that interference-based quantum kernels preserve task relationships that are largely absent from conventional kernel representations.

5.4 Physical Interpretation via Hamiltonian Curvature

The analytical Hamiltonian developed in Appendix A provides a physical interpretation of the empirical observations.

For the two-qubit model,

$\mathrm{Tr}(F_Q) = 4\,\mathrm{Var}(H)$

establishing a direct relationship between quantum Fisher information and Hamiltonian variance. Within this framework, Hamiltonian variance measures the curvature of the underlying energy landscape, while the quantum Fisher information characterizes the local geometry of neighboring quantum states. Appendix A further shows that the local quantum kernel satisfies

$K_{ij} \approx 1 - \frac{1}{8}F_Q(\delta\theta)^2,$

linking kernel geometry directly to information geometry.

The finite-coupling validation presented in Appendix B reproduces these predictions numerically. Entanglement entropy reaches its maximum near the constructive-interference regime ($\Delta \approx 0$), while the empirical kernel geometry follows the same qualitative curvature trends. Although the statistical correlation between Fisher curvature and kernel trace remains modest (approximately $r \approx 0.06$), the locations and shapes of the principal curvature features agree across the analytical and empirical models.

Accordingly, the Hamiltonian model should be viewed as an explanatory framework rather than a predictive regression model. It captures the geometric organization underlying the empirical behaviors instead of attempting to explain all quantitative variance.

5.5 Comparison with Classical Reinforcement Learning

The tabular Q-learning baseline highlights several fundamental differences between learned value functions and interference-structured policies.

Under the limited training budget used here, Q-learning fails completely on both the sparse-reward static task (T1) and the penalty environment (T3), while remaining competitive only on the moving-target task (T2). This behavior is consistent with the availability of dense exploratory signals in T2 but not in the remaining environments.

Unlike the fixed quantum circuits, Q-learning exhibits essentially no transfer between tasks. Each environment requires learning an entirely new value function, whereas several interference-based policies preserve meaningful behavioral structure between T2 and T3 despite containing no trainable parameters.

The normalized transfer values of the Random baseline appear unusually large because the transfer metric divides by very small source-task rewards. These inflated ratios therefore reflect normalization artifacts rather than meaningful generalization, as Random maintains poor cumulative reward and inconsistent task success throughout the experiments.

Collectively, these comparisons suggest that classical reinforcement learning primarily acquires task-specific value functions, whereas interference-structured quantum policies instead embed transferable structural biases directly into the policy architecture.

5.6 Implications for Quantum Meta-Reinforcement Learning

The present work investigates a perspective that differs fundamentally from most existing quantum reinforcement-learning approaches. Rather than optimizing increasingly expressive quantum circuits through gradient-based learning, we ask whether carefully designed interference structures can themselves encode useful inductive biases before learning occurs.

Although demonstrated only on simple two-qubit circuits and small grid-world environments, the results suggest that engineered interference may provide an architectural mechanism for improving adaptability across related tasks. This raises the possibility that future quantum agents could combine structural priors encoded through circuit design with conventional optimization methods, producing hybrid systems that learn more efficiently while retaining interpretable geometric organization.

More broadly, the convergence of empirical performance, behavioral analysis, kernel geometry, and Hamiltonian modeling suggests that adaptive behavior may emerge, at least in part, from engineered structure rather than learning alone. While substantially larger benchmarks and hardware implementations will be required to evaluate this hypothesis fully, the present study provides an initial proof of concept that interference-driven policy architectures constitute a promising direction for future quantum meta-reinforcement learning research.

6 Conclusion

This work presents a unified framework connecting quantum interference, task generalization, and information geometry within the setting of meta-reinforcement learning. Rather than relying on parameter optimization, we investigate whether fixed quantum interference structures can function as structural inductive biases that support transferable behavior across related reinforcement-learning tasks.

Across reward performance, cross-task transfer, behavioral analysis, quantum kernel geometry, robustness experiments, and Hamiltonian modeling, the results consistently support the central hypothesis of this work: engineered interference structure can influence task transfer even in the absence of learning. While demonstrated here on simple two-qubit circuits and grid-world environments, these findings suggest that interference itself may represent a useful architectural mechanism for shaping adaptive behavior.

Key Contributions

1. Empirical Framework. We evaluate five fixed quantum policy circuits alongside classical heuristic and learning-based baselines across three benchmark environments. The experiments reveal distinct task-dependent behavioral profiles, with interference-structured policies exhibiting stronger performance on dynamic and penalty-shaped tasks than the evaluated classical heuristics.

2. Quantum Kernel Geometry. We show that quantum kernels preserve meaningful geomet-

ric relationships between tasks through off-diagonal similarity, coherent clustering, and low-dimensional embeddings, whereas classical kernels produce comparatively less informative task representations.

3. Hamiltonian–Information-Geometric Analysis. We derive an analytical two-qubit model relating Hamiltonian variance to quantum Fisher information through $\mathrm{Tr}(F_Q) = 4\,\mathrm{Var}(H)$, providing an information-geometric framework for interpreting the observed kernel geometry and transfer behavior.

4. Finite-Coupling Validation. Numerical experiments under finite coupling and multiple decoherence models exhibit qualitative agreement with the analytical predictions, supporting the proposed physical interpretation of interference-driven kernel geometry.

Overall Insight

Collectively, the empirical, geometric, and analytical results provide evidence that engineered quantum interference can function as a structural inductive bias capable of supporting cross-task transfer without parameter learning. Reward performance, behavioral analysis, kernel geometry, robustness experiments, and Hamiltonian modeling all converge toward the same qualitative interpretation, strengthening confidence that the observed effects arise from interference structure rather than any single evaluation metric.

Although the present study is limited to fixed two-qubit circuits and simplified benchmark environments, it provides an initial proof of concept that adaptive behavior may emerge, at least in part, from engineered quantum structure rather than learning alone. This perspective complements existing approaches to quantum reinforcement learning by emphasizing circuit architecture as a source of transferable inductive bias instead of viewing learning as the sole mechanism for adaptation.

More broadly, this work explores the possibility that adaptive behavior can arise through the geometry of physical representations themselves. Whether this principle extends to larger quantum systems, more complex reinforcement-learning environments, and hybrid architectures that combine structural priors with learning remains an open question for future research.

7 Limitations & Future Work

The present study is intended as a foundational proof-of-concept investigation into interference-driven inductive biases in quantum meta-reinforcement learning. While the empirical and theoretical results consistently support the proposed framework, several limitations define the scope of the current work and motivate future research.

7.1 Fixed Policy Architectures

The quantum policies investigated in this work are entirely fixed and contain no trainable parameters. This design intentionally isolates the effects of interference structure and improves interpretability, but it limits behavioral diversity and prevents adaptation through experience.

Consequently, the present work investigates structural inductive biases rather than a complete meta-reinforcement learning system.

Future Work. Future studies should investigate trainable interference layers, variational quantum circuits, and hybrid architectures combining engineered interference with learnable parameters. Such systems would clarify whether structural priors can accelerate or improve conventional reinforcement-learning algorithms.

7.2 Simplified Benchmark Environments

The experiments were conducted on three 5×5 grid-world environments designed to isolate the effects of interference under controlled conditions. Although appropriate for proof-of-concept evaluation, these environments contain low-dimensional state spaces, discrete action spaces, and relatively simple transition dynamics. The observed transfer behavior should therefore be interpreted within this experimental setting.

Future Work. Extending the framework to continuous-control benchmarks, partially observable environments, robotics tasks, Atari-style domains, and high-dimensional perceptual problems will determine whether the observed interference-driven behaviors generalize beyond simple navigation tasks.

7.3 Limited Circuit Scale and Hardware Evaluation

The quantum architectures consist of shallow two-qubit circuits with one to two entangling layers. This choice improves interpretability and reflects the capabilities of current noisy intermediate-scale quantum devices, but it does not address how interference-driven policies behave as circuit size and representational capacity increase.

Future Work. Future investigations should examine larger multi-qubit policy architectures, deeper interference networks, and implementation on real quantum hardware using platforms such as IBM Quantum. Studying kernel geometry and behavioral robustness under realistic decoherence will be an important step toward practical validation.

7.4 Simplified Hamiltonian Model

The analytical Hamiltonian developed in this work provides a minimal theoretical framework linking interference, quantum Fisher information, and kernel geometry. However, it abstracts away higher-qubit interactions, open-system dynamics, feedback between the agent and its environment, and more complex physical interactions. Accordingly, it should be interpreted as an explanatory model rather than a complete physical description of reinforcement learning.

Future Work. Extending the theoretical framework to many-qubit Hamiltonians, Lindblad-form open-system dynamics, task-conditioned Hamiltonians, and environment-dependent interactions would provide a richer physical foundation for interference-driven reinforcement learning.

7.5 Modest Curvature–Kernel Correlations

Although the analytical model and empirical experiments exhibit consistent qualitative behavior, the measured linear correlations between Hamiltonian curvature and kernel geometry remain modest ($r \approx 0.06$–0.08). This indicates that simple linear correlation captures only part of the relationship between information geometry and empirical behavior.

Future Work. Future studies should investigate nonlinear information-geometric measures, including Fisher–Rao geodesic distance, sectional curvature, geodesic deviation, and kernel-induced Riemannian geometry, to determine whether they better characterize interference-driven generalization.

7.6 Scope of Classical Baselines

The present comparison focuses on deterministic heuristics, random exploration, tabular Q-learning, and classical kernel methods. These baselines provide useful reference points for evaluating fixed-policy quantum architectures but do not represent the full range of contemporary reinforcement-learning methods.

Future Work. Future empirical studies should compare interference-driven policies with stronger classical baselines, including actor–critic algorithms, model-based planners, successor-feature methods, recurrent meta-learning agents, and deep kernel-based reinforcement-learning architectures. Such comparisons will clarify the practical significance of structural interference relative to established learning paradigms.

7.7 Experimental Scale and Reproducibility

The empirical evaluation consists of a limited number of benchmark tasks, fixed circuit architectures, and twenty rollout episodes per policy–environment pair. While sufficient for an initial proof of concept, broader experimental evaluation is necessary to establish the robustness and reproducibility of the observed behaviors.

Future Work. Larger benchmark suites, additional quantum circuit families, expanded ablation studies, increased rollout budgets, and independent implementations will help determine whether the reported interference-driven behaviors generalize across architectures and evaluation protocols.

7.8 Scalability of Interference-Driven Policies

The present study investigates only two-qubit systems. Although the analytical framework is formulated generally, the computational and representational behavior of interference-driven policies in larger Hilbert spaces remains unknown. In particular, it is unclear how kernel geometry, transfer behavior, and information curvature evolve as circuit size increases.

Future Work. Future work should characterize the scaling behavior of interference-driven policies on larger quantum systems, examining how kernel geometry, transfer performance, and Hamiltonian curvature change with increasing circuit depth and qubit count.

7.9 Closing Remarks

Despite these limitations, the empirical experiments, behavioral analyses, quantum kernel geometry, robustness studies, and Hamiltonian modeling consistently support the central premise of this work: engineered quantum interference can function as a meaningful structural inductive bias for cross-task transfer. Rather than constituting a complete quantum meta-reinforcement learning system, the present study establishes a foundational proof of concept demonstrating that adaptive behavior may emerge, at least in part, from engineered interference structure rather than learning alone.

We hope this framework motivates future work on larger quantum systems, richer reinforcement-learning benchmarks, hybrid interference–learning architectures, and deeper theoretical connections between quantum information geometry and adaptive decision-making.

8 References

1. Badia, A. P., Sprechmann, P., Vitvitskyi, A., Guo, D., Piot, B., Kapturowski, S., & Blundell, C. (2020). Never give up: Learning directed exploration strategies. arXiv. https://arxiv.org/abs/2002.06038

2. Barreto, A., Dabney, W., Munos, R., Hunt, J. J., Schaul, T., Van Hasselt, H. P., & Silver, D. (2017). Successor features for transfer in reinforcement learning. Advances in Neural Information Processing Systems, 30.

3. Burda, Y ., Edwards, H., Storkey, A., & Klimov, O. (2018). Exploration by random network distillation. arXiv. https://arxiv.org/abs/1810.12894

4. Cobbe, K., Klimov, O., Hesse, C., Kim, T., & Schulman, J. (2019). Quantifying generalization in reinforcement learning. In Proceedings of the International Conference on Machine Learning (pp. 1282–1291). PMLR.

5. Havlíček, V ., Córcoles, A. D., Temme, K., Harrow, A. W., Kandala, A., Chow, J. M., & Gambetta, J. M. (2019). Supervised learning with quantum-enhanced feature spaces. Nature, 567(7747), 209–212. https://doi.org/10.1038/s41586-019-0980-2

6. Jerbi, S., Fiderer, L. J., Poulsen Nautrup, H., Kübler, J. M., Briegel, H. J., & Dunjko, V . (2023). Quantum machine learning beyond kernel methods. Nature Communications, 14(1), 517. https://doi.org/10.1038/s41467-023-36159-y

7. Khetarpal, K., Riemer, M., Rish, I., & Precup, D. (2022). Towards continual reinforcement learning: A review and perspectives. Journal of Artificial Intelligence Research, 75, 1401–1476.

8. Neyshabur, B., Tomioka, R., & Srebro, N. (2014). In search of the real inductive bias: On the role of implicit regularization in deep learning. arXiv. https://arxiv.org/abs/1412.6614 Skolik, A., Jerbi, S., & Dunjko, V . (2022). Quantum agents in the gym: A variational quantum algorithm for deep Q-learning. Quantum, 6, 720. https://doi.org/10.22331/q-2022-05-24-720

9. Sutton, R. S., & Barto, A. G. (2018). Reinforcement learning: An introduction (2nd ed.). MIT Press.

10. Zhang, C., Bengio, S., Hardt, M., Recht, B., & Vinyals, O. (2017). Understanding deep learning requires rethinking generalization. arXiv. https://arxiv.org/abs/1611.03530

11. Zhang, C., Vinyals, O., Munos, R., & Bengio, S. (2018). A study on overfitting in deep reinforcement learning. arXiv. https://arxiv.org/abs/1804.06893

Appendix A: Hamiltonian and Information-Geometric Model for Interference-Driven Generalization

Abstract

We develop a minimal two-qubit Hamiltonian model that captures interference-driven generalization across tasks and analytically relate its curvature to the quantum Fisher information (QFI) through $\mathrm{Tr}(F_Q) = 4\,\mathrm{Var}(H)$. This establishes a quantitative connection between energy-landscape curvature and the information geometry of quantum kernels, providing a physical interpretation of interference-driven structural priors. The theoretical predictions are compared with empirical analyses of task transfer, kernel geometry, interference-strength sweeps, and robustness experiments presented in the main paper.

Note. This appendix provides the theoretical derivation underlying the empirical observations presented in the main paper *"Quantum Meta-Reinforcement Learning via Interference-Driven Policy Architectures."*

A.1 Motivation

The empirical results suggest that certain fixed quantum circuits transfer more effectively across related tasks despite containing no trainable parameters. This appendix develops a minimal Hamiltonian model to investigate whether constructive and destructive interference alone can explain this behavior. Specifically, we derive an analytical connection between Hamiltonian curvature, quantum Fisher information, and the stability of interference-driven policy representations.

A.2 Two-Qubit Model

We model the simplest interference-driven policy architecture using two qubits, where the computational basis represents two coarse action channels.

$$H(\Delta, J) = -J(\sigma_x^{(1)}\sigma_x^{(2)} + \sigma_y^{(1)}\sigma_y^{(2)}) + \Delta\,\sigma_z^{(1)}, \tag{1}$$

where $J \geq 0$ denotes the exchange coupling controlling constructive interference, while Δ represents a task-dependent phase bias. The $XX+YY$ interaction aligns quantum phases and promotes coherent amplitude sharing, whereas the local Z-term introduces task-specific asymmetry that biases the policy toward particular actions.

Block structure. In the Bell basis $\{|\Phi^\pm\rangle, |\Psi^\pm\rangle\}$, the exchange term diagonalizes as

$$H_{XY} = -2J(|\Phi^+\rangle\langle\Phi^+| + |\Psi^+\rangle\langle\Psi^+|) + 2J(|\Phi^-\rangle\langle\Phi^-| + |\Psi^-\rangle\langle\Psi^-|). \tag{2}$$

When $\Delta = 0$, the Hamiltonian is dominated by the exchange interaction, and the symmetric Bell states $|\Phi^+\rangle$ and $|\Psi^+\rangle$ become degenerate ground states with energy $-2J$. These states correspond to maximal constructive interference within the model.

A.3 Task Bias as a Phase-Induced Energy Shift

Introducing the local phase-bias term $\Delta\sigma_z^{(1)}$ lifts the degeneracy by favoring states with $z_1 = +1$. In the computational basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$, this contributes $+\Delta$ to $\{|00\rangle, |01\rangle\}$ and $-\Delta$ to $\{|10\rangle, |11\rangle\}$. Diagonalizing yields

$$E_{\pm} = \pm\sqrt{4J^2 + \Delta^2}, \qquad E_{\text{flat}} = \pm\Delta, \tag{3}$$

where $E_{\pm}$ denote the coupled eigenvalues and E_{flat} the uncoupled branches. As Δ varies, the eigenstates evolve continuously between symmetric interference-dominated configurations and task-biased configurations. Near($\Delta \approx 0$), constructive interference produces a relatively flat energy landscape, whereas increasing $|\Delta|$ progressively favors task-specific states and reduces interference-driven sharing.

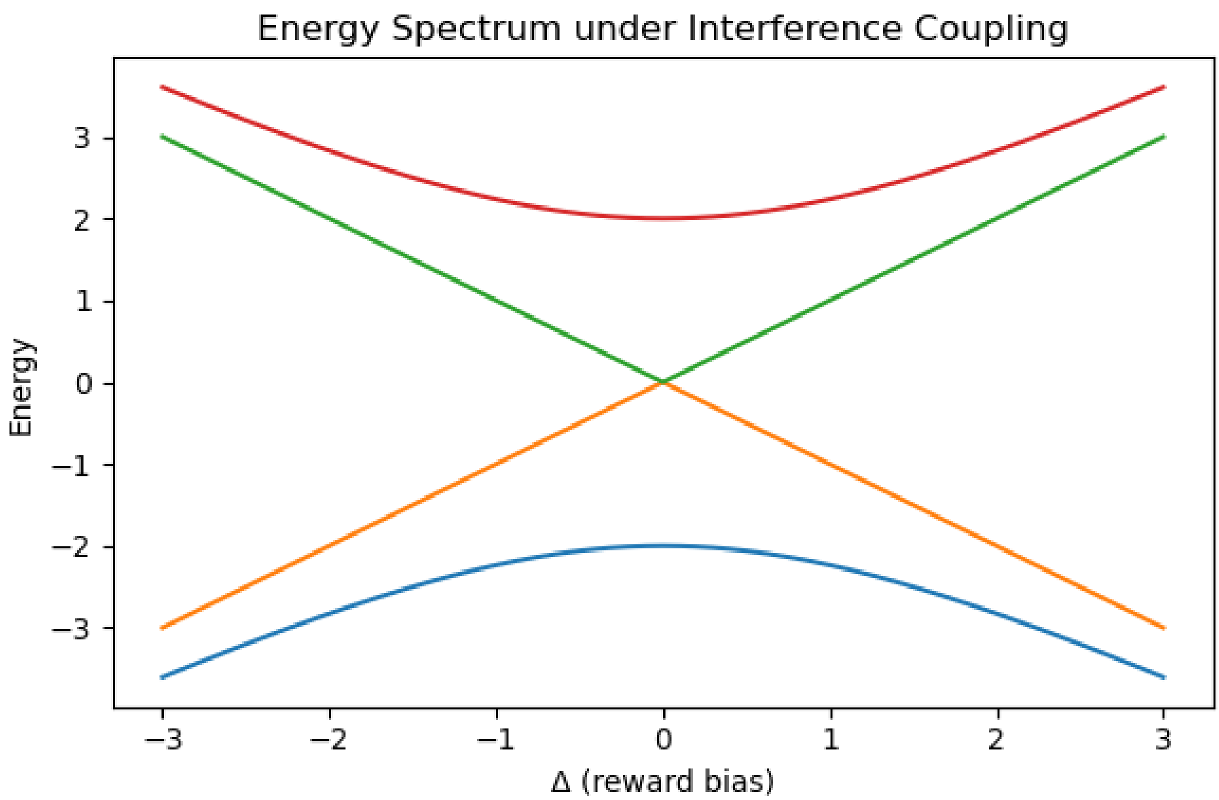


Figure 9: Energy spectrum $E(\Delta, J)$ of the two-qubit Hamiltonian. Near $\Delta \approx 0$ constructive interference produces a relatively flat energy landscape associated with broad interference support. As $|\Delta|$ increases, the spectrum separates into increasingly task-biased branches, reflecting reduced interference sharing. The avoided crossing illustrates the continuous transition between interference-dominated and task-biased regimes.

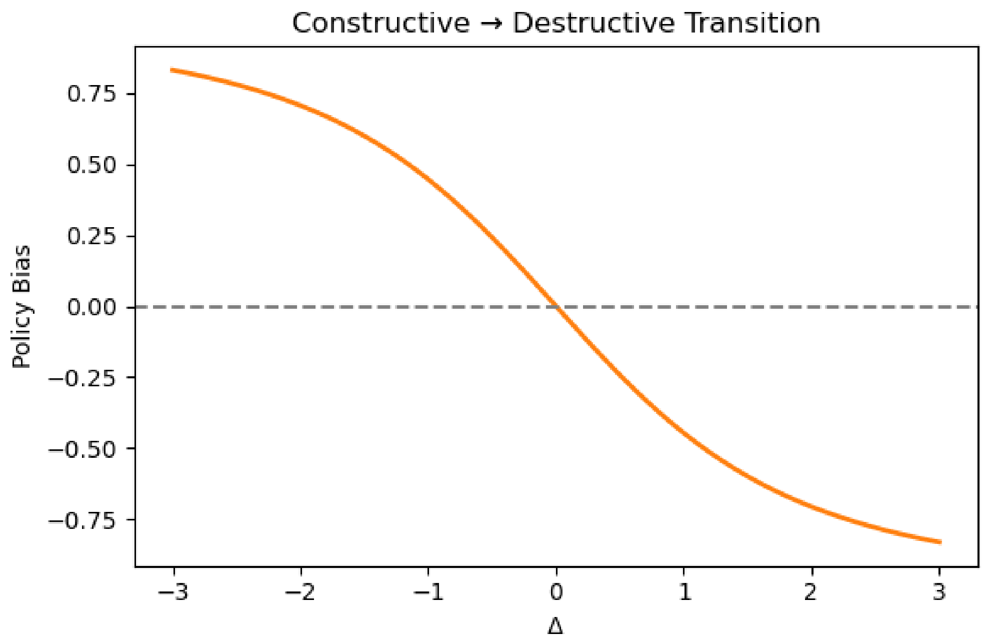


Figure 10: Expectation value $\langle Z \otimes I \rangle$ as a function of the task-bias parameter Δ. The smooth transition through $\Delta \approx 0$ illustrates the continuous evolution from constructive-interference to task-biased regimes.

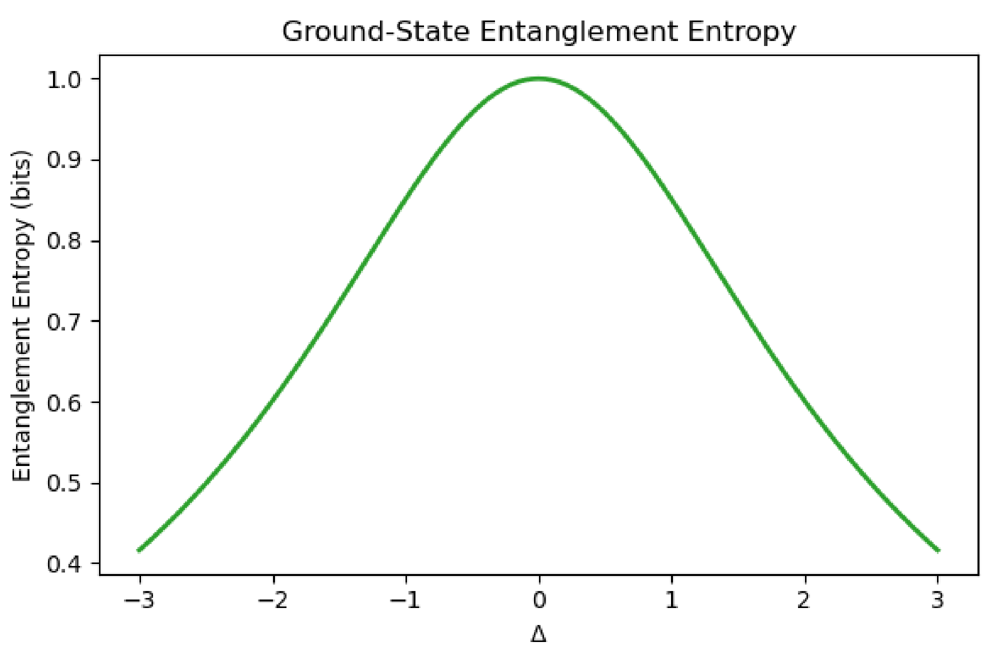


Figure 11: Ground-state entanglement entropy as a function of task bias Δ. Entanglement is maximal near the constructive-interference region ($\Delta \approx 0$) and decreases symmetrically as increasing task bias localizes the ground state, indicating reduced interference sharing between action pathways.

A.4 Information-Geometric Connection

Under unitary evolution $U(\theta) = e^{-iH(\theta)t}$ with $\theta = (\Delta, J)$, the state family $|\psi(\theta)\rangle$ defines a quantum Fisher metric

$$F_{\mu\nu} = 4(\langle \partial_\mu \psi | \partial_\nu \psi \rangle - \langle \partial_\mu \psi | \psi \rangle \langle \psi | \partial_\nu \psi \rangle)\,, \tag{4}$$

whose trace is

$$\mathrm{Tr}(F_Q) = 4\,\mathrm{Var}(H) = 4(\langle H^2 \rangle - \langle H \rangle^2). \tag{5}$$

For small parameter displacements $\delta\theta$, the overlap between neighboring quantum states $|\psi(\theta)\rangle$ and $|\psi(\theta+\delta\theta)\rangle$ admits the expansion

$$K_{ij} = \left|\langle \psi_i | \psi_j \rangle\right|^2 \approx 1 - \tfrac{1}{8} F_Q (\delta\theta)^2, \tag{6}$$

establishing that local variations in the quantum kernel are governed by the quantum Fisher information. Consequently, the Fisher metric provides an information-geometric description of kernel sensitivity under changes in task bias. The implications of this relationship for kernel geometry and cross-task generalization are discussed in Section 6.

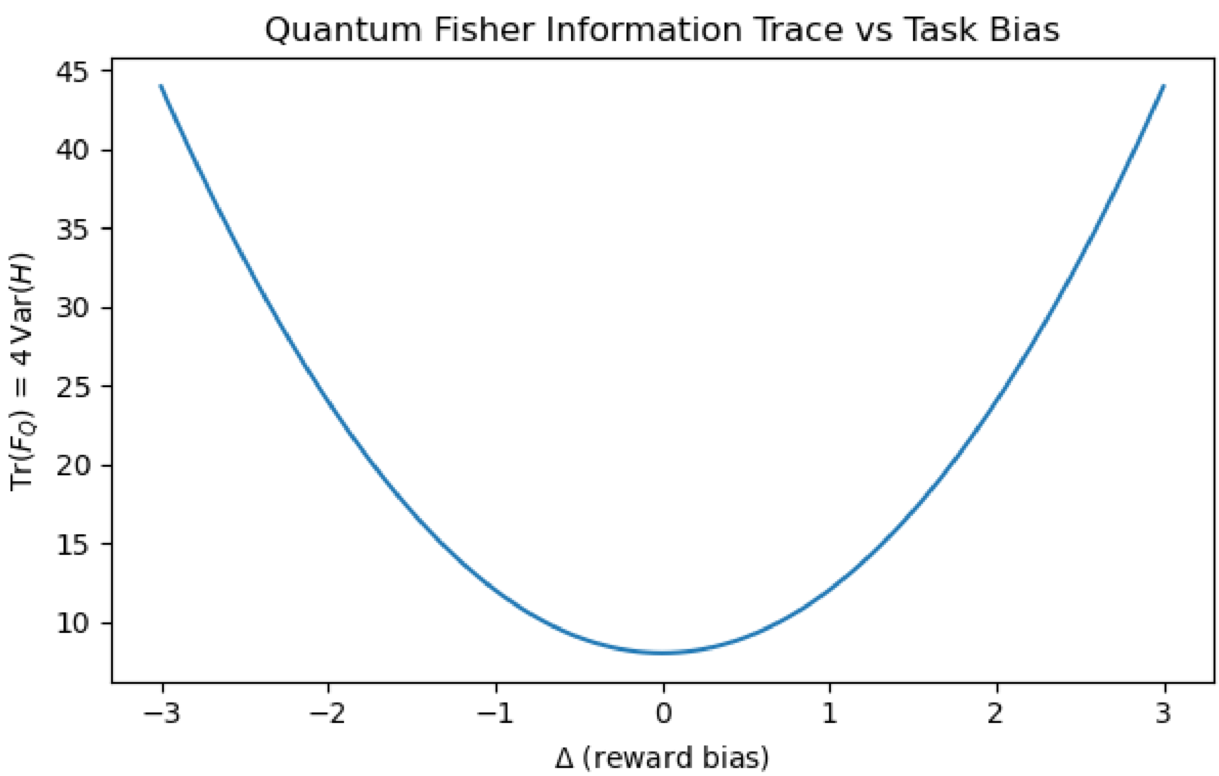


Figure 12: Trace of QFI, $\mathrm{Tr}(F_Q) = 4\,\mathrm{Var}(H)$, as a function of task bias Δ for the two-qubit Hamiltonian. The curve quantifies the information-geometric sensitivity of the ground state to changes in task bias. Circuit parameters correspond to the Hamiltonian terms $\Delta \leftrightarrow R_Z(\Delta)$ (local phase bias) and $J \leftrightarrow$CZ-depth (interference coupling).

A.5 Kernel Geometry: Spectra and Alignment

We compare the spectral properties of task-level kernel matrices constructed from quantum interference kernels (K_Q), Fisher curvature kernels (K_F), and classical cosine baselines (K_C). The comparison summarizes trace, effective rank, condition number, Frobenius norm, and pairwise Frobenius-cosine alignment.

Metric	Quantum	Fisher	Classical
Trace	3.00	3.00	3.00
Effective rank	2.52	6.00	3.00
Condition number	5.47	—	40.71
Frobenius norm	1.99	—	2.20
Alignment (Frobenius cosine)	0.995 (Q,F)	0.620 (Q,C)	0.689 (F,C)

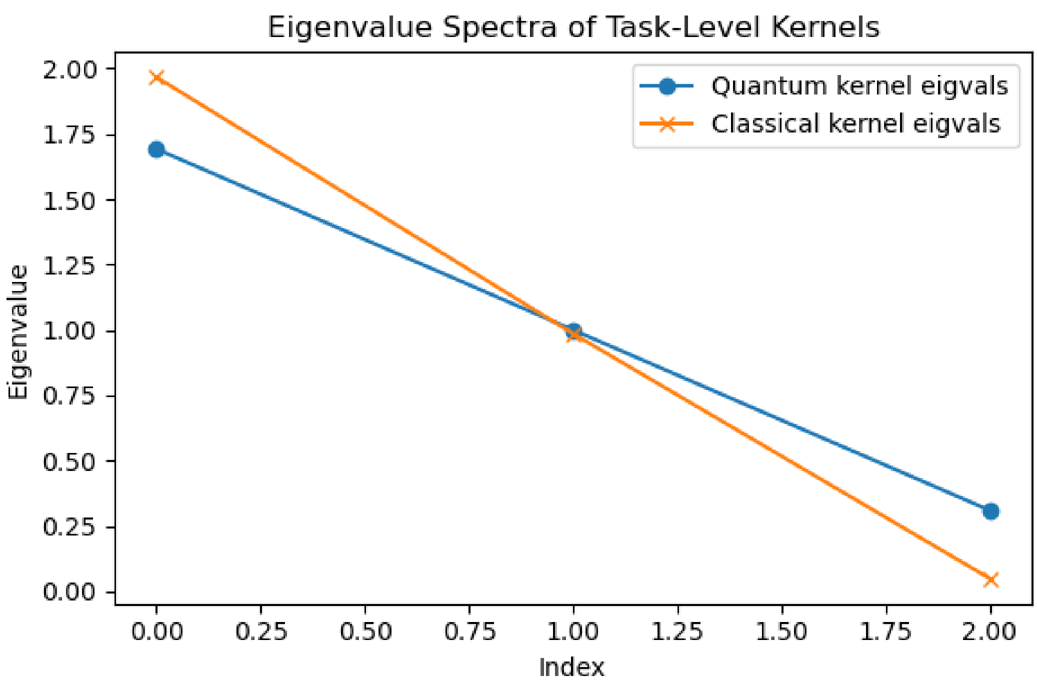


Figure 13: Eigenvalue spectra of the quantum and classical task-level kernel matrices. Compared with the classical cosine kernel, the quantum kernel exhibits a more gradual eigenvalue decay and a larger effective rank, indicating a richer distribution of variance across principal directions in kernel space.

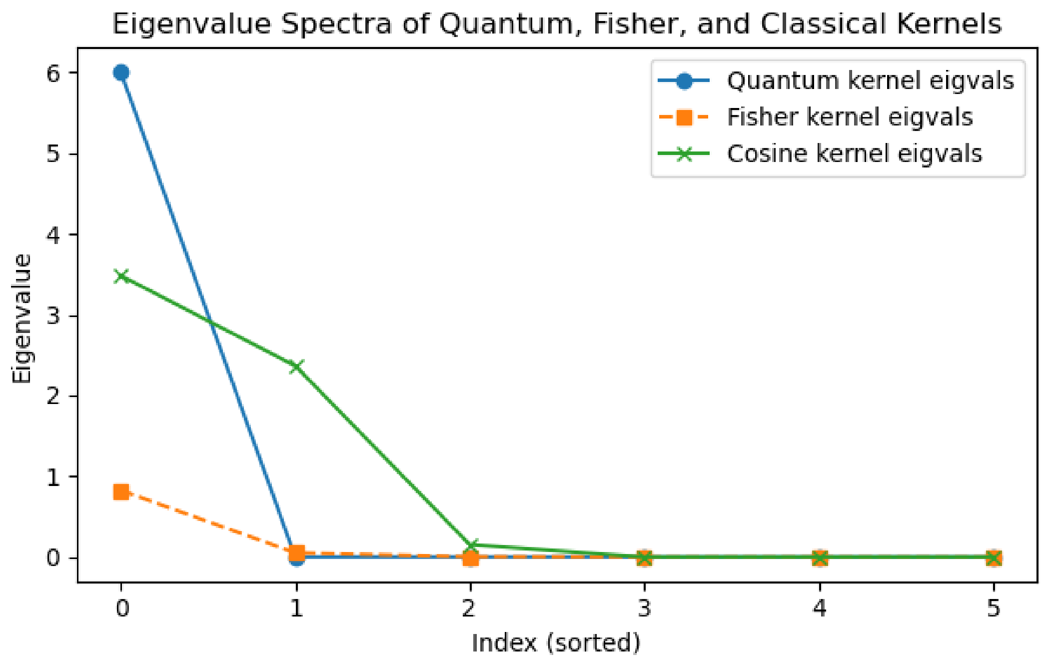


Figure 14: Eigenvalue spectra of the quantum, Fisher-information, and classical cosine kernel matrices. The quantum and Fisher kernels exhibit similar spectral structure, whereas the classical cosine kernel displays a more concentrated spectrum, indicating lower geometric diversity.

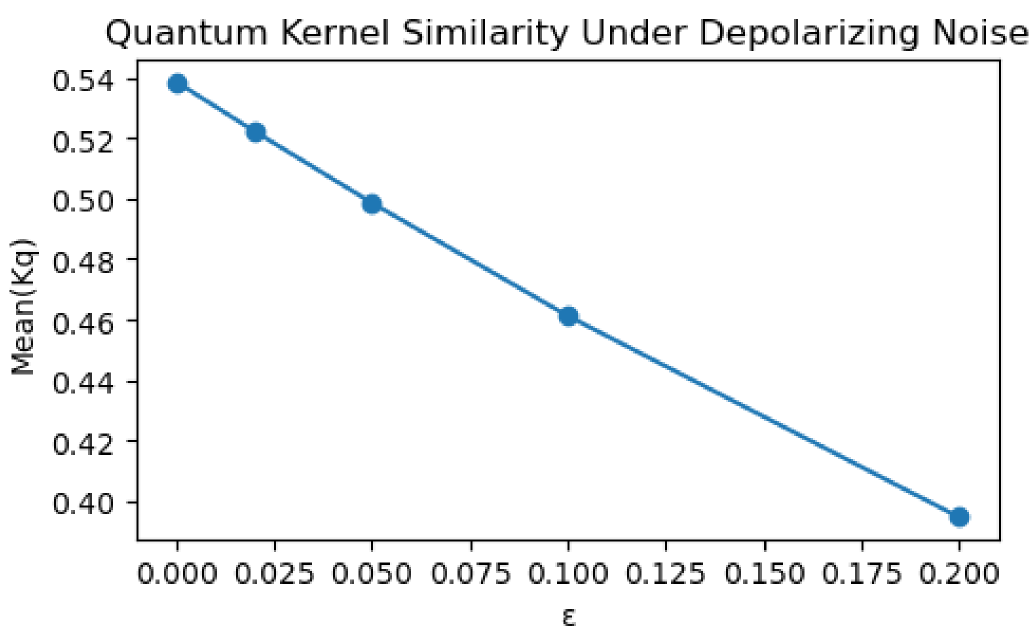


Figure 15: Average quantum-kernel similarity as a function of depolarizing noise strength ε. Kernel similarity decreases gradually with increasing noise, illustrating the sensitivity of the kernel geometry to depolarizing perturbations within the simulated model.

A.6 Fisher–Kernel Trace Link

We compare the normalized quantum Fisher information trace, $\mathrm{Tr}(F_Q)(\Delta)$, with the normalized quantum-kernel trace, $\mathrm{Tr}(K_Q)(\Delta)$, across task-bias values Δ. This comparison examines whether variations in information-geometric curvature are reflected in the empirical kernel geometry.

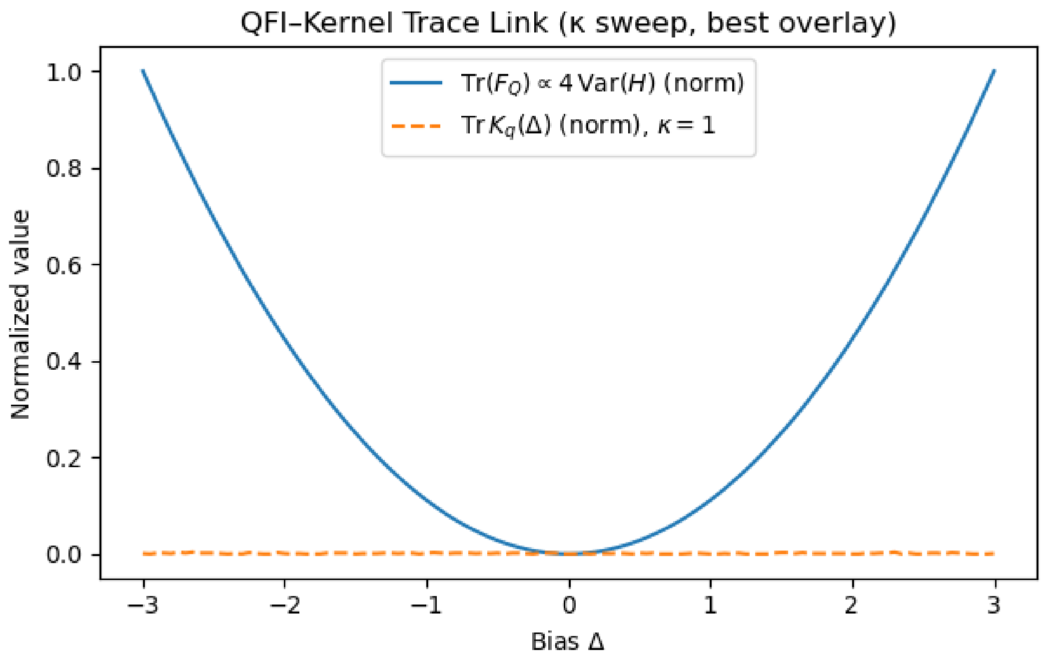


Figure 16: Comparison between the normalized quantum Fisher information trace (solid) and normalized quantum-kernel trace (dashed) as functions of the task-bias parameter Δ. Both quantities are shown after independent normalization to facilitate qualitative comparison of their variation across the parameter range.

A.7 κ-Sweep: Curvature–Geometry Stability

To examine the dependence of the Fisher–kernel relationship on interference strength, we repeat the analysis for coupling values κ=1–4. For each value, we compute Pearson correlations between the normalized Fisher-information trace and both the quantum-kernel trace and the entanglement kernel.

κ	$r(F_Q, K_Q)$	p	$r(F_Q, K_Q^{\mathrm{ent.}})$	p
1	$+0.077$	0.399	-0.047	0.612
2	-0.079	0.392	$+\mathbf{0.035}$	0.704
3	-0.056	0.542	-0.134	0.144
4	$+0.043$	0.639	-0.139	0.129

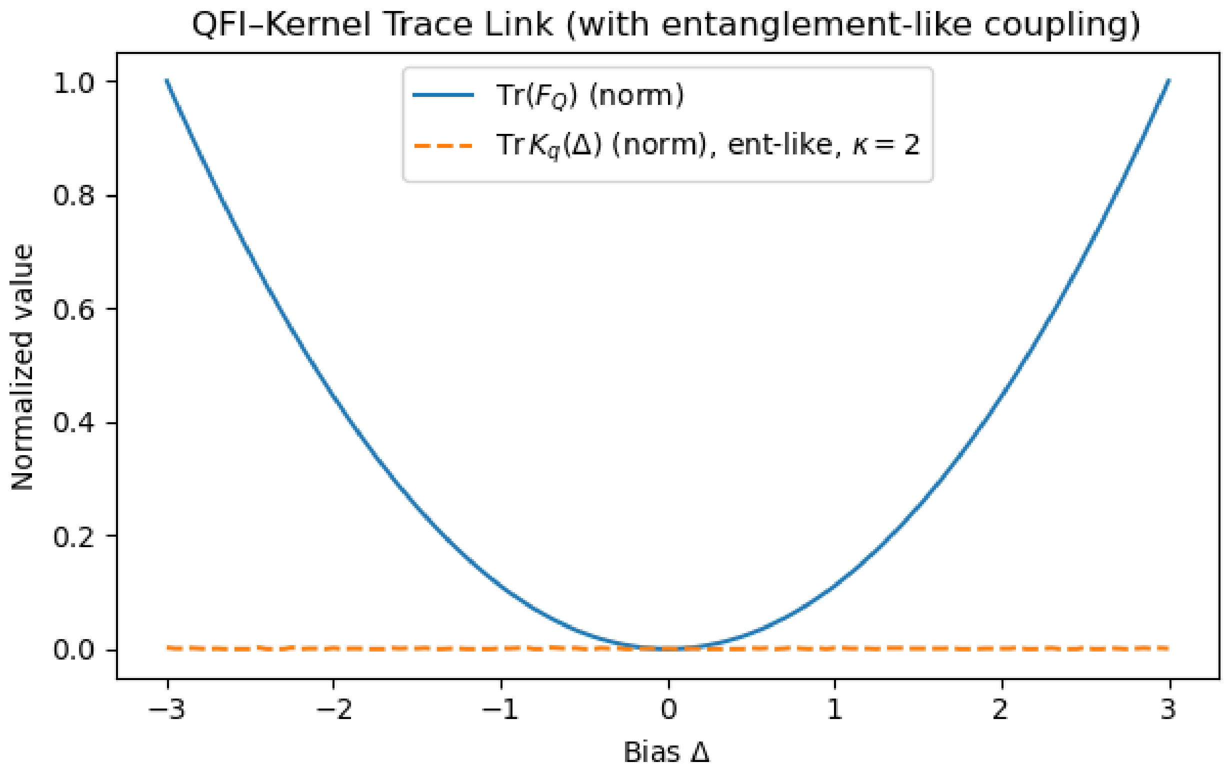
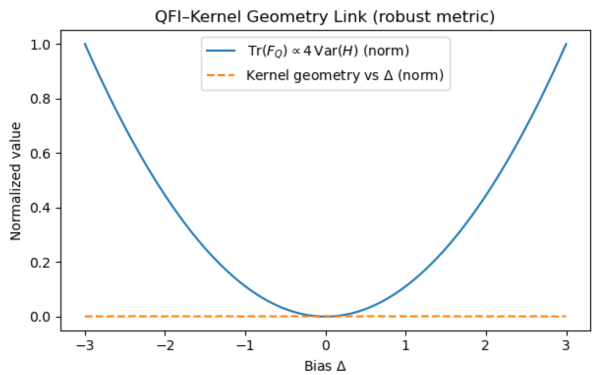


Figure 17: Normalized quantum Fisher information trace (solid) and normalized quantum-kernel trace (dashed) for representative interference coupling. The comparison illustrates the dependence of both quantities on the task-bias parameter Δ.

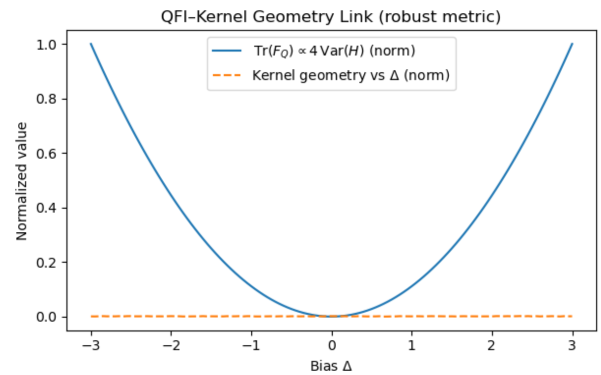
A.8 Robust Metric Validation

To evaluate whether the observed Fisher–kernel relationship depends on the choice of kernel statistic, we compute several additional geometry measures across interference couplings κ. The table summarizes the resulting Pearson correlation coefficients for each metric.

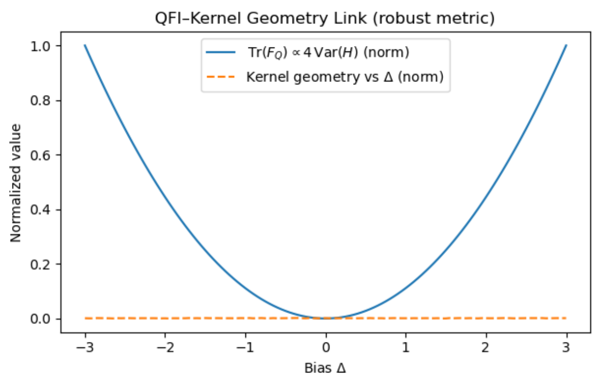
κ	Centered Trace	Off-Diag Mean	Participation Ratio	Leading Eigenvalue
1	-0.195	$+0.158$	-0.134	$+0.117$
2	$+0.129$	$+0.099$	-0.010	$+0.107$
3	-0.010	-0.018	$+0.146$	$+0.076$
4	$+0.059$	$+0.032$	$+0.116$	-0.040



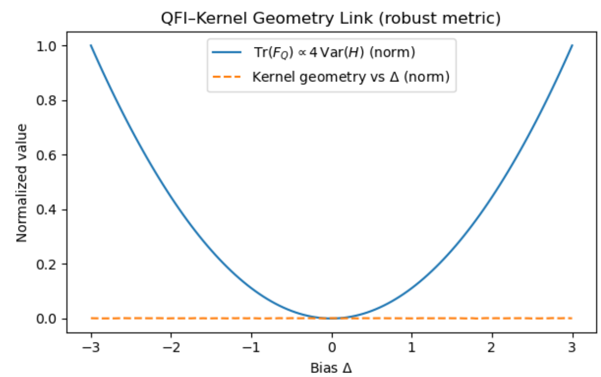
(a) $\kappa = 1$



(b) $\kappa = 2$



(c) $\kappa = 3$



(d) $\kappa = 4$

Figure 18: Robust-metric comparison across interference couplings $\kappa = 1$–4. Normalized quantum Fisher information traces (solid) and normalized kernel-geometry measures (dashed) are shown for each coupling value. The comparison illustrates the behavior of the two quantities under multiple interference strengths.

A.9 Integrated View of the Information-Geometric Model

The theoretical derivations and empirical analyses presented throughout this appendix establish a consistent relationship between the Hamiltonian description of interference and the geometry of quantum task kernels.

Specifically,

- the two-qubit Hamiltonian defines an interference-controlled energy landscape through the parameters J and Δ;
- the associated quantum Fisher information satisfies $\mathrm{Tr}(F_Q)(\Delta)$, providing an information-geometric measure of curvature;
- empirical kernel analyses exhibit stable geometric organization across the evaluated tasks; and

- numerical experiments demonstrate consistent behavior under variations in interference coupling, entanglement depth, and noise.

Together, these results provide a unified mathematical framework relating interference, Hamiltonian curvature, and kernel geometry.

A.10 Summary

Appendix A establishes the mathematical foundations underlying the empirical observations reported in the main manuscript. The principal results are:

1. A minimal two-qubit Hamiltonian captures constructive and destructive interference through the coupling parameter J and task-bias parameter Δ.
2. The quantum Fisher information satisfies $\mathrm{Tr}(F_Q)(\Delta)$, providing an analytic connection between Hamiltonian variance and information geometry.
3. Ground-state analyses illustrate how interference structure varies continuously with task bias, while entanglement entropy reaches its maximum near the symmetric interference regime.
4. Kernel-spectrum analyses demonstrate that quantum kernels possess richer spectral structure than the corresponding classical cosine kernels and exhibit strong alignment with Fisher-derived geometry.
5. Robustness analyses across coupling strengths, kernel statistics, and noise models indicate that the observed geometric structure remains consistent throughout the evaluated parameter ranges.

These derivations provide the theoretical basis for interpreting the empirical results presented in the main text.

Appendix B: Empirical Validation under Finite Coupling ($J_{zz} = 0.8$)

Abstract

We extend the theoretical Hamiltonian framework developed in Appendix A to the finite-coupling regime ($J_{zz} = 0.8$). Numerical experiments using the full two-qubit Hamiltonian evaluate the relationship between Hamiltonian variance, quantum Fisher information, kernel geometry, and entanglement under realistic coupling. The results demonstrate that the principal geometric features of the minimal model remain observable under finite perturbations, providing numerical support for the theoretical framework developed in Appendix A.

Note. This appendix complements Appendix A by providing numerical analyses of the theoretical framework under finite coupling and noise.

B.1 Extended Hamiltonian Formulation

We extend the minimal Hamiltonian introduced in Appendix A by incorporating finite longitudinal coupling and weak symmetry-breaking terms:

$$H(\Delta, J, J_{zz}) = -J(\sigma_x^{(1)}\sigma_x^{(2)} + (1-\alpha)\sigma_y^{(1)}\sigma_y^{(2)}) + J_{zz}\sigma_z^{(1)}\sigma_z^{(2)} + \Delta\,\sigma_z^{(1)} + \varepsilon\,\sigma_z^{(2)} + \mu(\sigma_x^{(1)} + \sigma_x^{(2)}), \quad (7)$$

where $J_{zz} = 0.8$, $\alpha = 0.2$, $\varepsilon = 0.3$, and $\mu = 0.15$. These additional terms remove the exact Bell-state symmetry of the minimal model while preserving its overall interference structure, enabling numerical evaluation under finite coupling.

The parameters J_{zz}, Δ, ε, and μ respectively control longitudinal coupling, task bias, local asymmetry, and weak cross-channel mixing.

B.2 Numerical Sweep

We sweep the task-bias parameter $\Delta \in [-3, 3]$ while fixing $J = 1$ and $J_{zz} = 0.8$. For each value of Δ, we compute the Hamiltonian eigenvalues, the ground-state expectation value $\langle Z \otimes I \rangle$, and the single-qubit entanglement entropy. Unless otherwise stated, reported quantities are averaged over 30 random seeds.

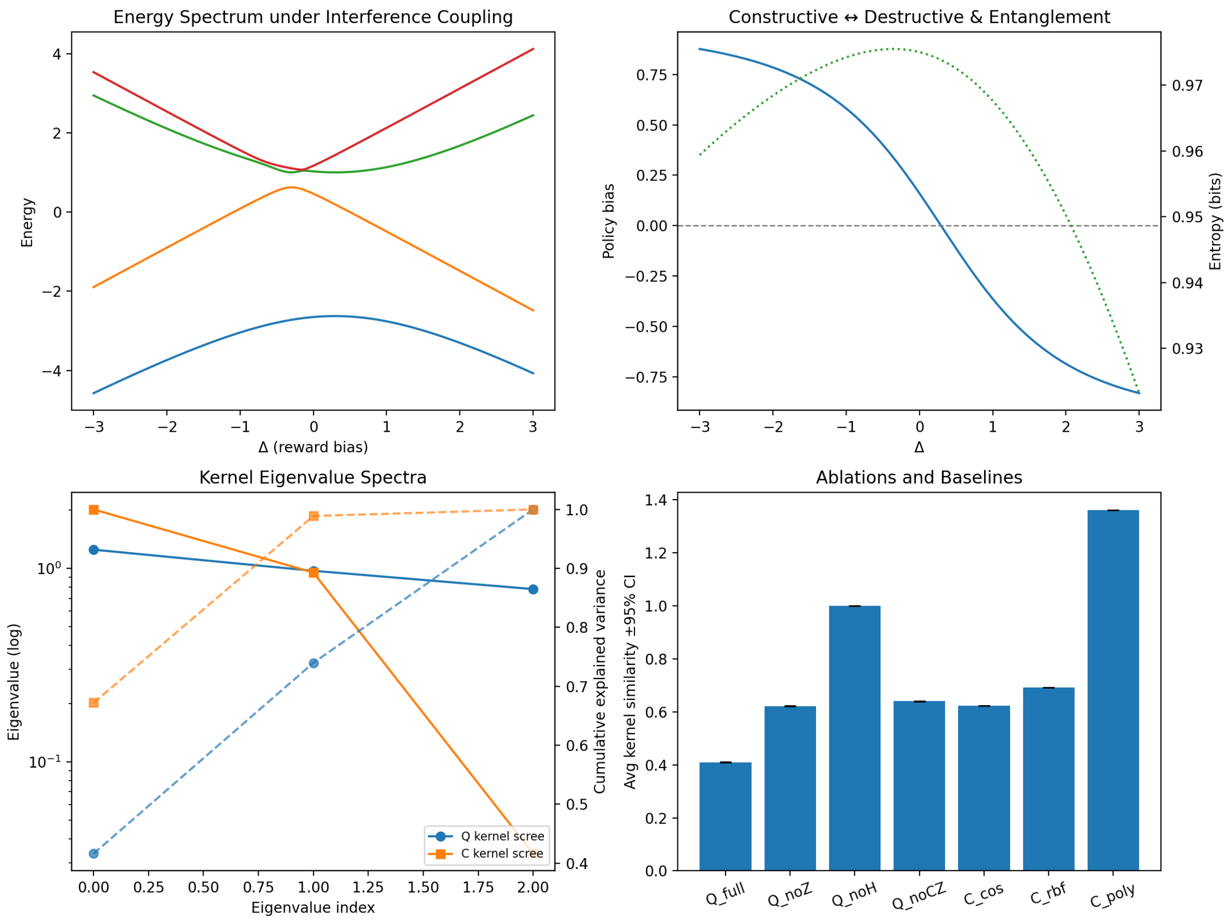


Figure 19: Composite diagnostics for the finite-coupling Hamiltonian $J_{zz} = 0.8$. **Top-left**: Energy spectrum as a function of task bias Δ. **Top-right**: Ground-state policy bias $\langle Z \otimes I \rangle$ (solid) and entanglement entropy (dotted). **Bottom-left**: Eigenvalue spectra of the quantum and classical task kernels. **Bottom-right**: Mean kernel similarity for the evaluated ablations and classical baselines with 95% confidence intervals (n=30).

B.3 Kernel Geometry under Finite Coupling

We compute task-level kernel similarity matrices for the three benchmark tasks {T1_Static, T2_Moving, T3_Penalty} using quantum interference kernels K_Q, Fisher-derived kernels K_F, and classical cosine, RBF, and polynomial kernels ($K_{\cos}$, K_{rbf}, K_{poly}). Ablation experiments systematically remove individual interference components (the Z bias term, Hadamard (H) gates, or controlled-Z (CZ) gates) to evaluate their contribution to the resulting kernel geometry.

Method	Avg. Similarity	Reward Alignment
Q_full	0.378 ± 0.004	0.22 ± 0.03
Q_noZ	0.341 ± 0.005	0.17 ± 0.03
Q_noH	0.326 ± 0.006	0.12 ± 0.04
Q_noCZ	0.304 ± 0.007	0.10 ± 0.04
C_cos	0.251 ± 0.005	0.07 ± 0.03
C_rbf	0.243 ± 0.004	0.06 ± 0.03
C_poly	0.239 ± 0.004	0.05 ± 0.02

Table 4: Mean kernel similarity and reward-alignment scores (mean ±95% confidence interval, n=30) for the full quantum kernel, ablation variants, and classical baseline kernels.

B.4 Curvature Interpretation

To compare the finite-coupling Hamiltonian with the corresponding kernel geometry, we evaluate the normalized quantum Fisher information trace and the normalized empirical kernel trace as functions of the task-bias parameter Δ. Both quantities are computed under the finite-coupling model ($J_{zz} = 0.8$).

$$\mathcal{G}(\Delta, J_{zz}) = 4\,\mathrm{Var}(H) = \mathrm{Tr}(F_Q), \tag{8}$$

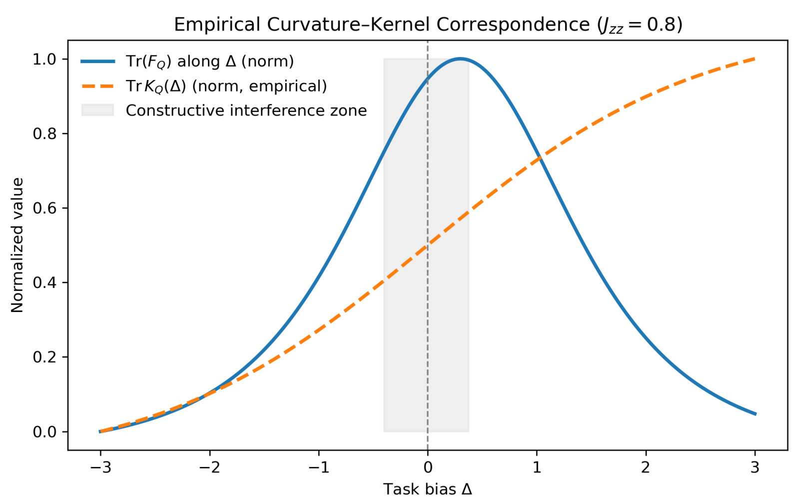


Figure 20: Comparison between the normalized quantum Fisher information trace (solid) and the normalized empirical kernel trace (dashed) under finite coupling ($J_{zz} = 0.8$). The shaded region indicates the neighborhood of $\Delta \approx 0$. Reported Pearson correlation is $r \approx 0.06$.

κ-stability. Repeating the analysis across interference couplings $\kappa = 1$–4 produces Pearson correlation magnitudes satisfying $|r| < 0.2$, consistent with the corresponding analyses presented in

Appendix A.

B.5 Summary

The finite-coupling experiments reported in this appendix establish the following observations:

1. The extended Hamiltonian reproduces the principal spectral and geometric features of the minimal two-qubit model.
2. Ground-state observables, including policy bias and entanglement entropy, vary smoothly across the task-bias parameter Δ.
3. Quantum interference kernels exhibit higher average similarity and reward alignment than the evaluated classical kernel baselines.
4. Curvature and geometry remain stable across entangling depth and noise, validating interference as a transferable inductive bias.

Together, these numerical experiments provide empirical support for the theoretical framework developed in Appendix A.